



*COMP7503 Multimedia Technologies*

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# Multimedia Documents

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THE UNIVERSITY OF HONG KONG

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D E P A R T M E N T O F  
**COMPUTER SCIENCE**

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# Preamble

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- ❖ We have learnt about a variety of sensors, for multiple stream of sensory data, in our earlier discussions
  - ❖ That means, we have discussed about the *multi* part
- ❖ In this topic, we shall go through important concepts that combine sensory data into a palatable and presentable format
  - ❖ As a result, we are going to talk about the *medium* part, which is usually referred to as *document*
- ❖ Any example documents?



# Example Documents

- ❖ Printed books
  - ❖ Mostly text based
- ❖ Newspaper
  - ❖ Text + Images
- ❖ Webpages
  - ❖ Text + Images
- ❖ Music File
  - ❖ Audio + Metadata
- ❖ Video File
  - ❖ Video + Audio + Metadata
- ❖ Electronic Text Documents
  - ❖ MSWord, Pages, PDF
  - ❖ PPT, Keynotes

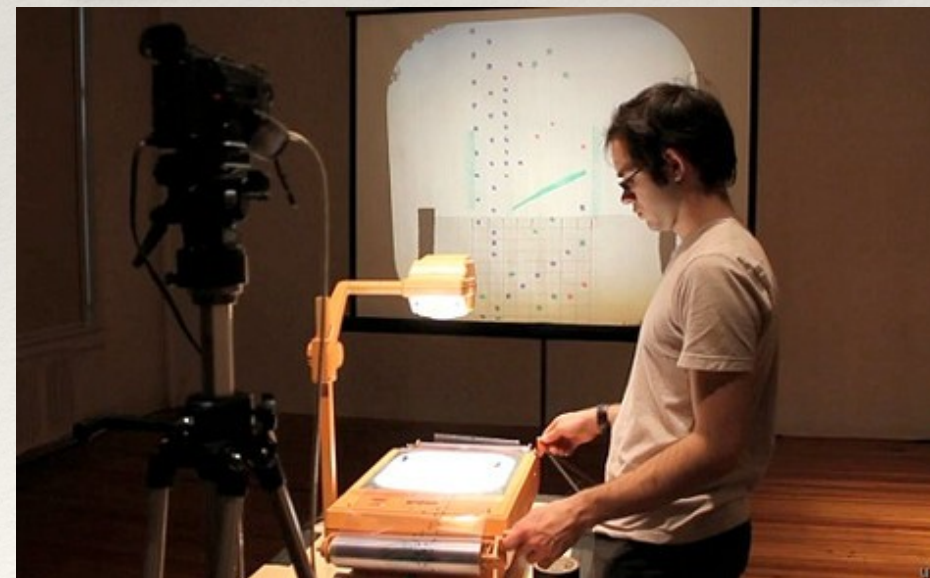
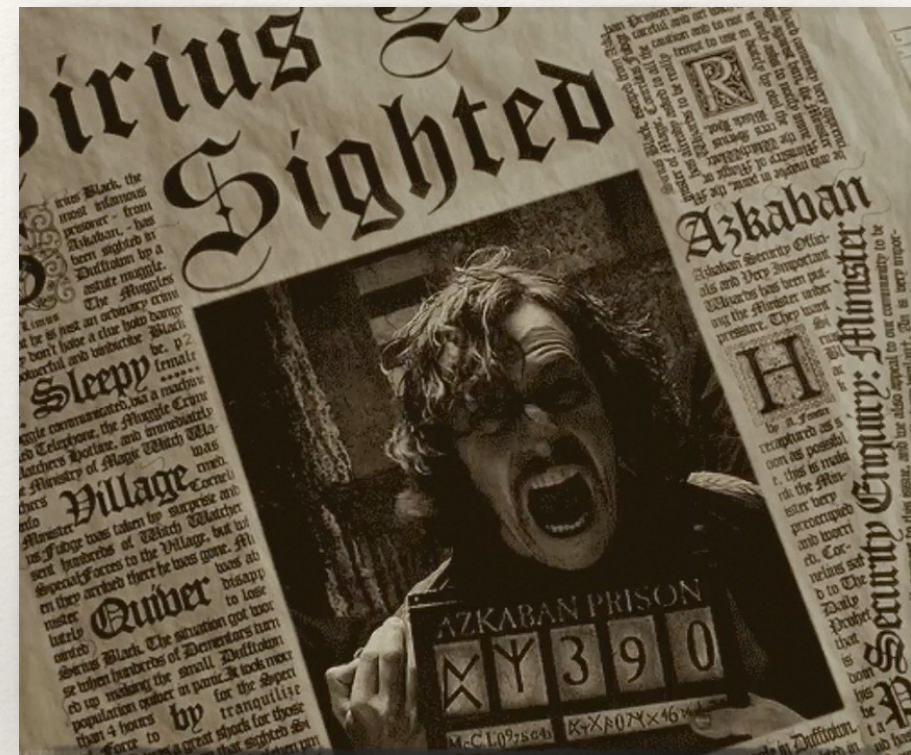


Image source: <http://www.ubergizmo.com/2010/05/color-a-sound-relies-on-projector-to-create-a-manual-jukebox/>



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# Book

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- ❖ A book is regarded as a collection of text and images presented by an author to make it
  - ❖ Coherent and Complete
  - ❖ Frozen in a particular time
- ❖ Any subsequent edition of the book is carefully thought about and prepared
  - ❖ To make it complete and to keep it current for the foreseeable future



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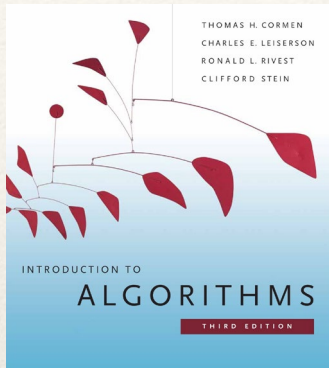
# Organization of a book

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- ❖ Elementary unit of a book
  - ❖ *Page*
- ❖ Multiple pages, covering related objects and events, are pooled together to form a *Chapter*
- ❖ Chapters sharing the same scope might be further pooled together to form *Parts*
- ❖ Multiple books
- ❖ Other considerations
  - ❖ Text and images should be reasonably sized for ease of reading
  - ❖ The size and weight of the whole book should allow a normal person to handle it properly



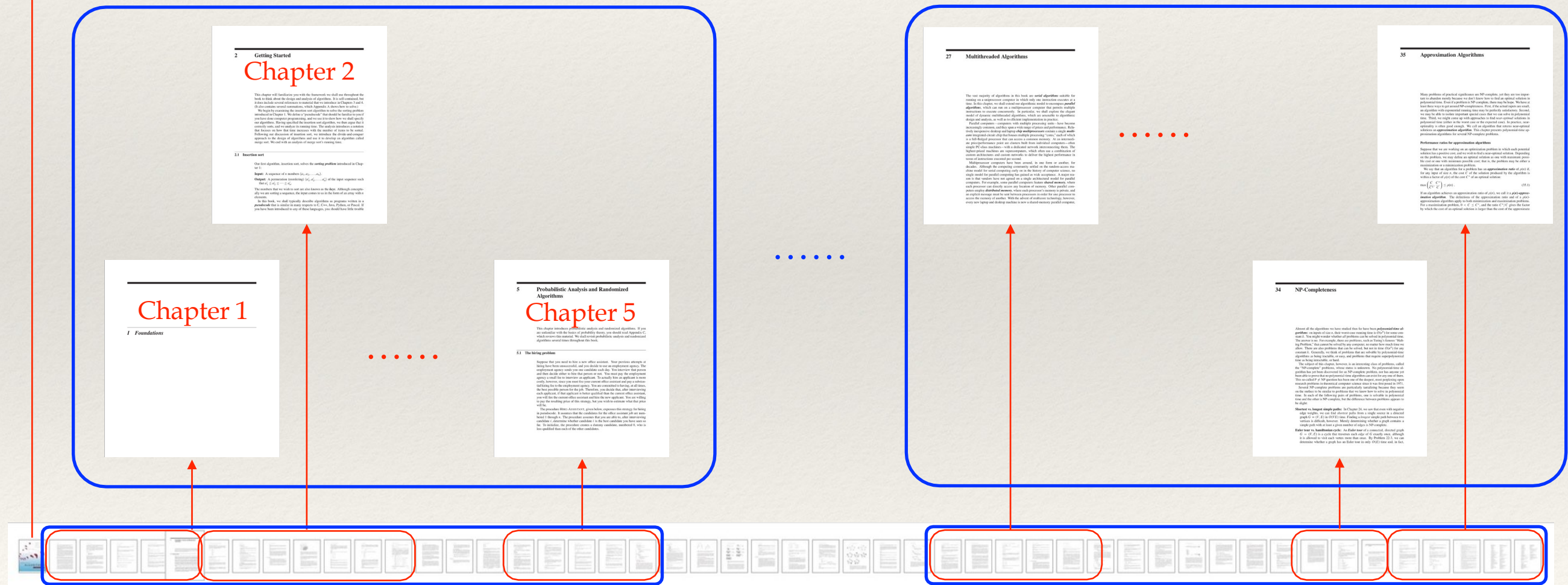
# Tree Structure of a Book



What do the texts in the book represent?

Part-I

Part-VII



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# Text in the Book

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- ❖ Text is a symbolic representation of speech
- ❖ Speech is a temporal signal
- ❖ A book is created by considering a timeline corresponding to the text describing the topic and dividing it into appropriate intervals
  - ❖ These intervals are converted into pages
  - ❖ Each page is a *folded version of the timeline*



# Text Style and Spatial Layout

> REPLACE THIS LINE WITH YOUR PAPER IDENTIFICATION NUMBER (DOUBLE-CLICK HERE TO EDIT) < 1

## Preparation of Papers for IEEE TRANSACTIONS and JOURNALS (December 2013)

First A. Author, *Fellow, IEEE*, Second B. Author, and Third C. Author, Jr., *Member, IEEE*

**Abstract**—These instructions give you guidelines for preparing papers for IEEE Transactions and Journals. Use this document as a template if you are using Microsoft *Word* 6.0 or later. Otherwise, use this document as an instruction set. The electronic file of your paper will be formatted further at IEEE. Paper titles should be written in uppercase and lowercase letters, not all uppercase. Avoid writing long formulas with subscripts in the title; short formulas that identify the elements are fine (e.g., “Nd–Fe–B”). Do not write “(Invited)” in the title. Full names of authors are preferred in the author field, but are not required. Put a space between authors’ initials. Define all symbols used in the abstract. Do not cite references in the abstract. Do not delete the blank line immediately above the abstract; it sets the footnote at the bottom of this column.

**Index Terms**—Enter key words or phrases in alphabetical order, separated by commas. For a list of suggested keywords, send a blank e-mail to [keywords@ieee.org](mailto:keywords@ieee.org) or visit [http://www.ieee.org/organizations/pubs/ani\\_prod/keywrd98.txt](http://www.ieee.org/organizations/pubs/ani_prod/keywrd98.txt)

### I. INTRODUCTION

THIS document is a template for Microsoft *Word* versions 6.0 or later. If you are reading a paper or PDF version of this document, please download the electronic file, TRANS-JOUR.DOC, from the IEEE Web site at <http://www.ieee.org/web/publications/authors/transjnl/index.html> so you can use it to prepare your manuscript. If you would prefer to use LATEX, download IEEE’s LATEX style and sample files from the same Web page. Use these LATEX files for formatting, but please follow the instructions in TRANS-JOUR.DOC or TRANS-JOUR.PDF.

If your paper is intended for a conference, please contact your conference editor concerning acceptable word processor formats for your particular conference.

This paragraph of the first footnote will contain the date on which you submitted your paper for review. It will also contain support information, including sponsor and financial support acknowledgment. For example, “This work was supported in part by the U.S. Department of Commerce under Grant BS123456”.

The next few paragraphs should contain the authors’ current affiliations, including current address and e-mail. For example, F. A. Author is with the National Institute of Standards and Technology, Boulder, CO 80305 USA (e-mail: [author@boulder.nist.gov](mailto:author@boulder.nist.gov)).

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### II. GUIDELINES FOR MANUSCRIPT PREPARATION

When you open TRANS-JOUR.DOC, select “Page Layout” from the “View” menu in the menu bar (View | Page Layout), (these instructions assume MS 6.0. Some versions may have alternate ways to access the same functionalities noted here). Then, type over sections of TRANS-JOUR.DOC or cut and paste from another document and use markup styles. The pull-down style menu is at the left of the Formatting Toolbar at the top of your *Word* window (for example, the style at this point in the document is “Text”). Highlight a section that you want to designate with a certain style, then select the appropriate name on the style menu. The style will adjust your fonts and line spacing. Do not change the font sizes or line spacing to squeeze more text into a limited number of pages. Use italics for emphasis; do not underline.

To insert images in *Word*, position the cursor at the insertion point and either use Insert | Picture | From File or copy the image to the Windows clipboard and then Edit | Paste Special | Picture (with “float over text” unchecked).

IEEE will do the final formatting of your paper. If your paper is intended for a conference, please observe the conference page limits.

#### A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have already been defined in the abstract. Abbreviations such as IEEE, SI, ac, and dc do not have to be defined. Abbreviations that incorporate periods should not have spaces: write “C.N.R.S.,” not “C. N. R. S.” Do not use abbreviations in the title unless they are unavoidable (for example, “IEEE” in the title of this article).

#### B. Other Recommendations

Use one space after periods and colons. Hyphenate complex modifiers: “zero-field-cooled magnetization.” Avoid dangling participles, such as, “Using (1), the potential was calculated.” [It is not clear who or what used (1).] Write instead, “The potential was calculated by using (1),” or “Using (1), we calculated the potential.”

Use a zero before decimal points: “0.25,” not “.25.” Use “cm<sup>3</sup>,” not “cc.” Indicate sample dimensions as “0.1 cm × 0.2 cm,” not “0.1 × 0.2 cm<sup>2</sup>.” The abbreviation for “seconds” is “s,” not “sec.” Use “Wb/m<sup>2</sup>” or “webers per square meter,” not “webers/m<sup>2</sup>.” When expressing a range of values, write “7 to 9” or “7–9,” not “7~9.”

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A parenthetical statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.) In American English, periods and commas are within quotation marks, like “this period.” Other punctuation is “outside”! Avoid contractions; for example, write “do not” instead of “don’t.” The serial comma is preferred: “A, B, and C” instead of “A, B and C.”

If you wish, you may write in the first person singular or plural and use the active voice (“I observed that ...” or “We observed that ...” instead of “It was observed that ...”). Remember to check spelling. If your native language is not English, please get a native English-speaking colleague to carefully proofread your paper.

#### C. How to Create a PostScript File

First, download a PostScript printer driver from <http://www.adobe.com/support/downloads/pdrrwin.htm> (for Windows) or from <http://www.adobe.com/support/downloads/pdrrvmac.htm> (for Macintosh) and install the “Generic PostScript Printer” definition. In *Word*, paste your figure into a new document. Print to a file using the PostScript printer driver. File names should be of the form “fig5.ps.” Use Open Type fonts when creating your figures, if possible. A listing of the acceptable fonts are as follows: Open Type Fonts: Times Roman, Helvetica, Helvetica Narrow, Courier, Symbol, Palatino, Avant Garde, Bookman, Zapf Chancery, Zapf Dingbats, and New Century Schoolbook.

### III. MATH

If you are using *Word*, use either the Microsoft Equation Editor or the *MathType* add-on (<http://www.mathtype.com>) for equations in your paper (Insert | Object | Create New | Microsoft Equation or *MathType* Equation). “Float over text” should *not* be selected.

#### A. Equations

Number equations consecutively with equation numbers in parentheses flush with the right margin, as in (1). First use the equation editor to create the equation. Then select the “Equation” markup style. Press the tab key and write the equation number in parentheses. To make your equations more compact, you may use the solidus ( / ), the exp function, or appropriate exponents. Use parentheses to avoid ambiguities in denominators. Punctuate equations when they are part of a sentence, as in

$$\int_0^{\tau_2} F(r, \varphi) dr d\varphi = [\sigma r_2 / (2\mu_0)] \cdot \int_0^{\infty} \exp(-\lambda |z_j - z_0|) \lambda^{-1} J(\lambda r) J(\lambda r) d\lambda. \quad (1)$$

Be sure that the symbols in your equation have been defined before the equation appears or immediately following.

Italicize symbols (*T* might refer to temperature, but *T* is the unit tesla). Refer to “(1),” not “Eq. (1)” or “equation (1),” except at the beginning of a sentence: “Equation (1) is ...”

### IV. UNITS

Use either SI (MKS) or CGS as primary units. (SI units are strongly encouraged.) English units may be used as secondary units (in parentheses). This applies to papers in data storage. For example, write “15 Gb/cm<sup>2</sup> (100 Gb/in<sup>2</sup>).” An exception is when English units are used as identifiers in trade, such as “3½-in disk drive.” Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity in an equation.

The SI unit for magnetic field strength *H* is A/m. However, if you wish to use units of *T*, either refer to magnetic flux density *B* or magnetic field strength symbolized as  $\mu_0 H$ . Use the center dot to separate compound units, e.g., “A·m<sup>2</sup>.”

### V. SOME COMMON MISTAKES

The word “data” is plural, not singular. The subscript for the permeability of vacuum  $\mu_0$  is zero, not a lowercase letter “o.” The term for residual magnetization is “remanence”; the adjective is “remanent”; do not write “remnance” or “remnant.” Use the word “micrometer” instead of “micron.” A graph within a graph is an “inset,” not an “insert.” The word “alternatively” is preferred to the word “alternately” (unless you really mean something that alternates). Use the word “whereas” instead of “while” (unless you are referring to simultaneous events). Do not use the word “essentially” to mean “approximately” or “effectively.” Do not use the word “issue” as a euphemism for “problem.” When compositions are not specified, separate chemical symbols by en-dashes; for example, “NiMn” indicates the intermetallic compound Ni<sub>0.5</sub>Mn<sub>0.5</sub> whereas “Ni–Mn” indicates an alloy of some composition Ni<sub>x</sub>Mn<sub>1-x</sub>.

Be aware of the different meanings of the homophones “affect” (usually a verb) and “effect” (usually a noun), “complement” and “compliment,” “discreet” and “discrete,” “principal” (e.g., “principal investigator”) and “principle” (e.g., “principle of measurement”). Do not confuse “imply” and “infer.”

Prefixes such as “non,” “sub,” “micro,” “multi,” and “ultra” are not independent words; they should be joined to the words they modify, usually without a hyphen. There is no period after the “et” in the Latin abbreviation “*et al.*” (it is also italicized). The abbreviation “i.e.,” means “that is,” and the abbreviation “e.g.,” means “for example” (these abbreviations are not italicized).

A general IEEE styleguide is available at <http://www.ieee.org/web/publications/authors/transjnl/index.html>

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# Linear Structure of Books

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- ❖ Paper-based physical representation forced linear structure on books
- ❖ Authors prepare a book assuming that it will be read linearly
  - ❖ Though, it is theoretically possible to read a book in any order, but most books, particularly fiction, are read in the linear order



# E-Pub Text

## 2 Multimedia: A Definition

Only a few inventions in the history of civilization have had the same impact on society in so many ways and at so many levels as computers. Where once we used computers for computing with simple alphanumeric data, we now use them primarily to exchange information, to communicate, and to share experiences. Computers are rapidly evolving as a means for gaining insights and sharing ideas across distance and time.

Multimedia computing started gaining serious attention from researchers and practitioners during the 1990s. Before 1991, people talked about multimedia, but the computing power, storage, bandwidth, and processing algorithms were not advanced enough to deal with audio and video. With the increasing availability and popularity of CDs, people became excited about creating documents that could include not only text, but also images, audio, and even video. That decade saw explosive growth in all aspects of hardware and software technology related to multimedia computing and communication. In the early 1990s, PC manufacturers labeled their high-end units containing advanced graphics *multimedia PCs*. That

Smaller Text

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Bigger text with different fonts



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# Transformation in documents

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- ❖ Audio and video technologies changed the book metaphor for documents
  - ❖ They allow rendering of time-varying audio and video signals in their natural audio and video form
  - ❖ No need to artificially represent time as visual pages
- ❖ Hyperlinks in electronic documents and the introduction on hyperlinked pages on web allow documents to be read in an order that a reader finds appropriate rather than what the author intended
- ❖ Now a document is no longer a compact and closed physical artifact, but a dynamically and organically growing result of collaborative authoring that presents multimedia in all its forms



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# Content Segments

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- ❖ Most documents may be considered a composition of many *content segments* (CS)
- ❖ A CS is a component that has been either authored or captured, which can be considered an independent unit, a.k.a atomic segment, of media that could be combined with other segments
- ❖ A CS could be
  - ❖ Text document
  - ❖ Photo
  - ❖ Video Segment
  - ❖ Audio Segment
  - ❖ or any other similar data that represents a particular media



# CS Examples

- Given several relevant CS for producing a document, it is possible to combine them in many different ways to result in different documents

Screen showing Demo of the system.  
CS1

Nice food.  
CS2

TV Advertisement that became very popular.  
CS3

Nice music piece.  
CS4

Terrace at Del Mar Plaza has nice place to Enjoy views.  
CS5

CS6

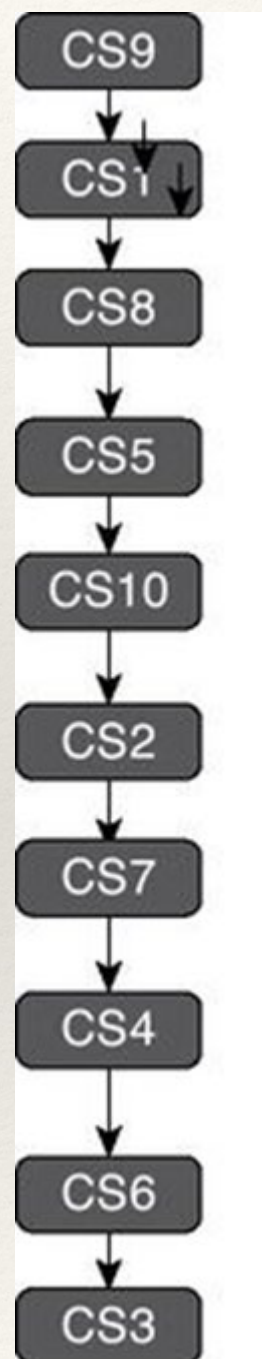
CS7

CS8

CS9

CS10

# Combining CS



Traditional  
Linear  
Segment



CS9

Screen showing  
Demo of the system.

CS1



CS10

Nice food.

CS2



CS6

TV  
Advertisement  
that became  
very popular.

CS3



CS8

Terrace at Del Mar  
Plaza has nice place to  
Enjoy views.

CS5



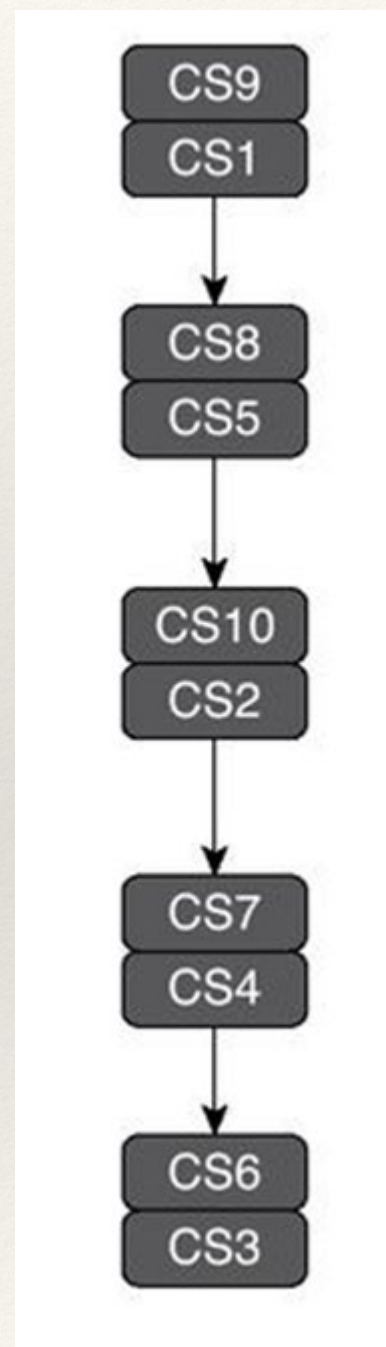
CS7

Nice music piece.

CS4



# Combining CS (Cont'd)



Screen showing  
Demo of the system.

CS1

CS9



Terrace at Del Mar  
Plaza has nice place to  
Enjoy views.

CS5

CS8



Nice food.

CS2

CS10



Nice music piece.

CS4

CS7



TV  
Advertisement  
that became  
very popular.

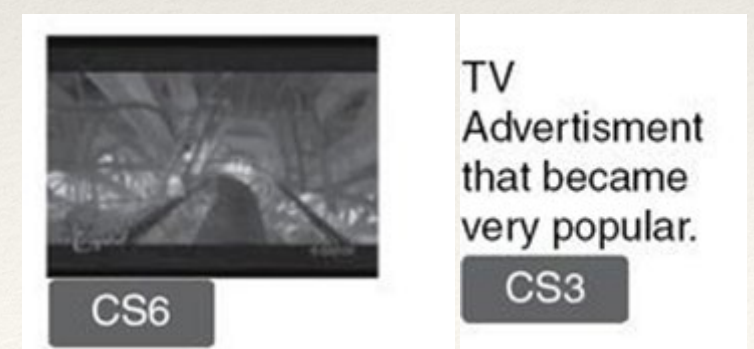
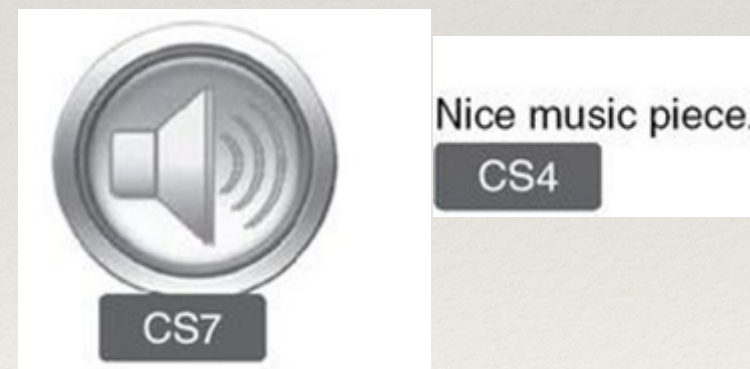
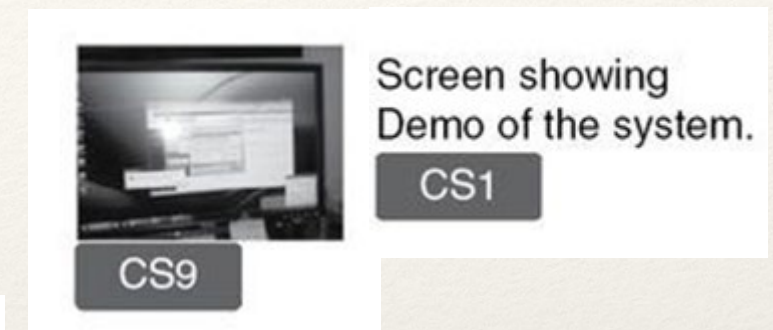
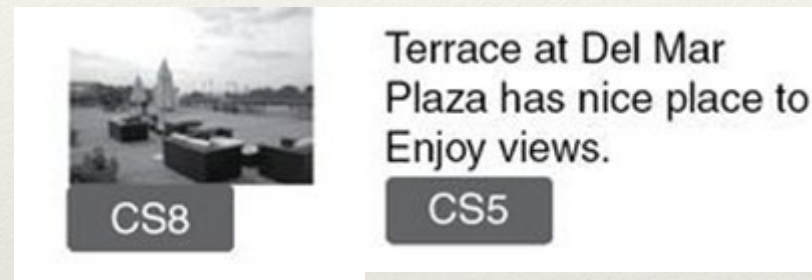
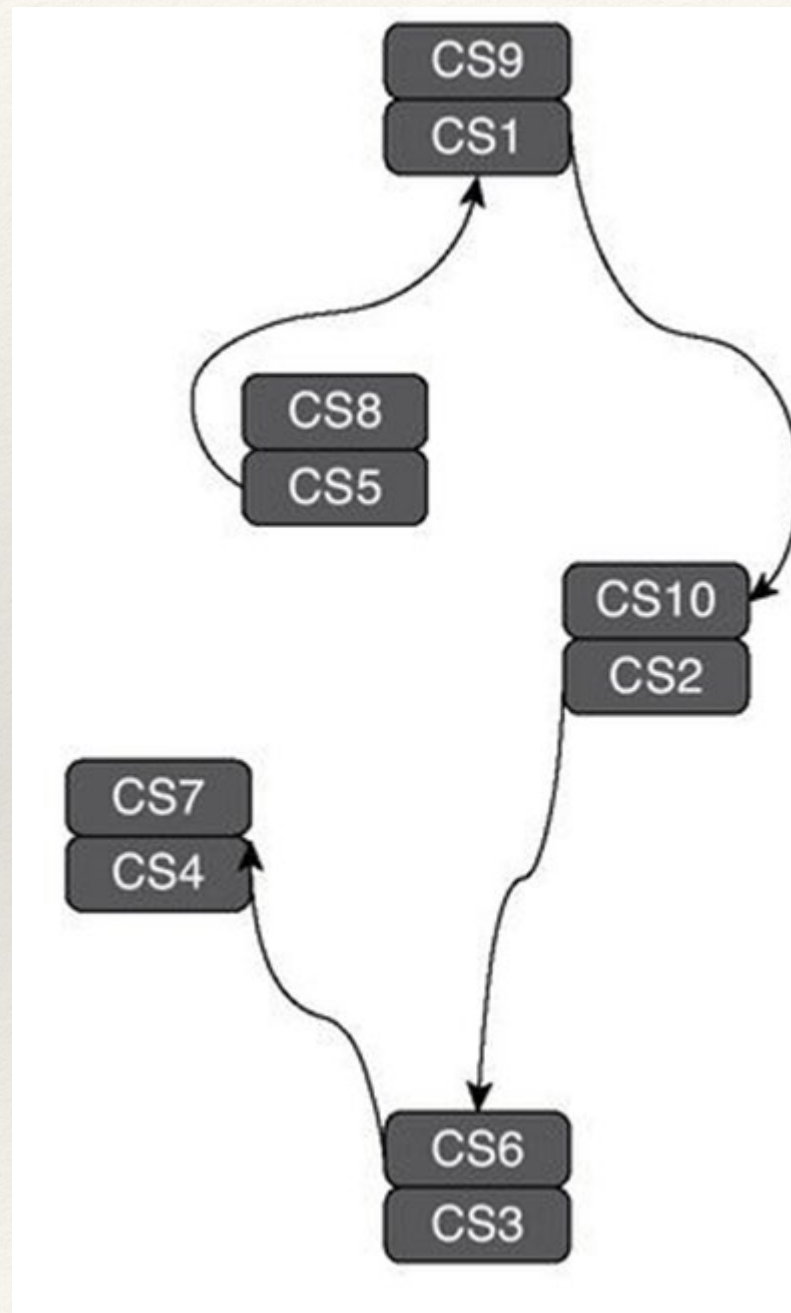
CS3

CS6

Atomic Segments  
Combined to form  
Composite Elements



# Combining CS (Cont'd)



Composite Elements are linked to others, and user can navigate in non-linear order

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# Evolution of Documents

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- ❖ Evolve along three dimensions

- ❖ *Type of Media*

- ❖ Text remains an important component of documents, but now documents use different media as the author sees fit
    - ❖ Author can combine different media in space and time to communicate his ideas in the most compelling manner
    - ❖ The same CS in electronic form can be used as many times as necessary

- ❖ *Non-Linear Flow*

- ❖ With the advent of electronic media and the ability to create links, the limitation of linearity can be easily overcome
    - ❖ This provides a very flexible method of authoring documents that may be customised for different types of audiences

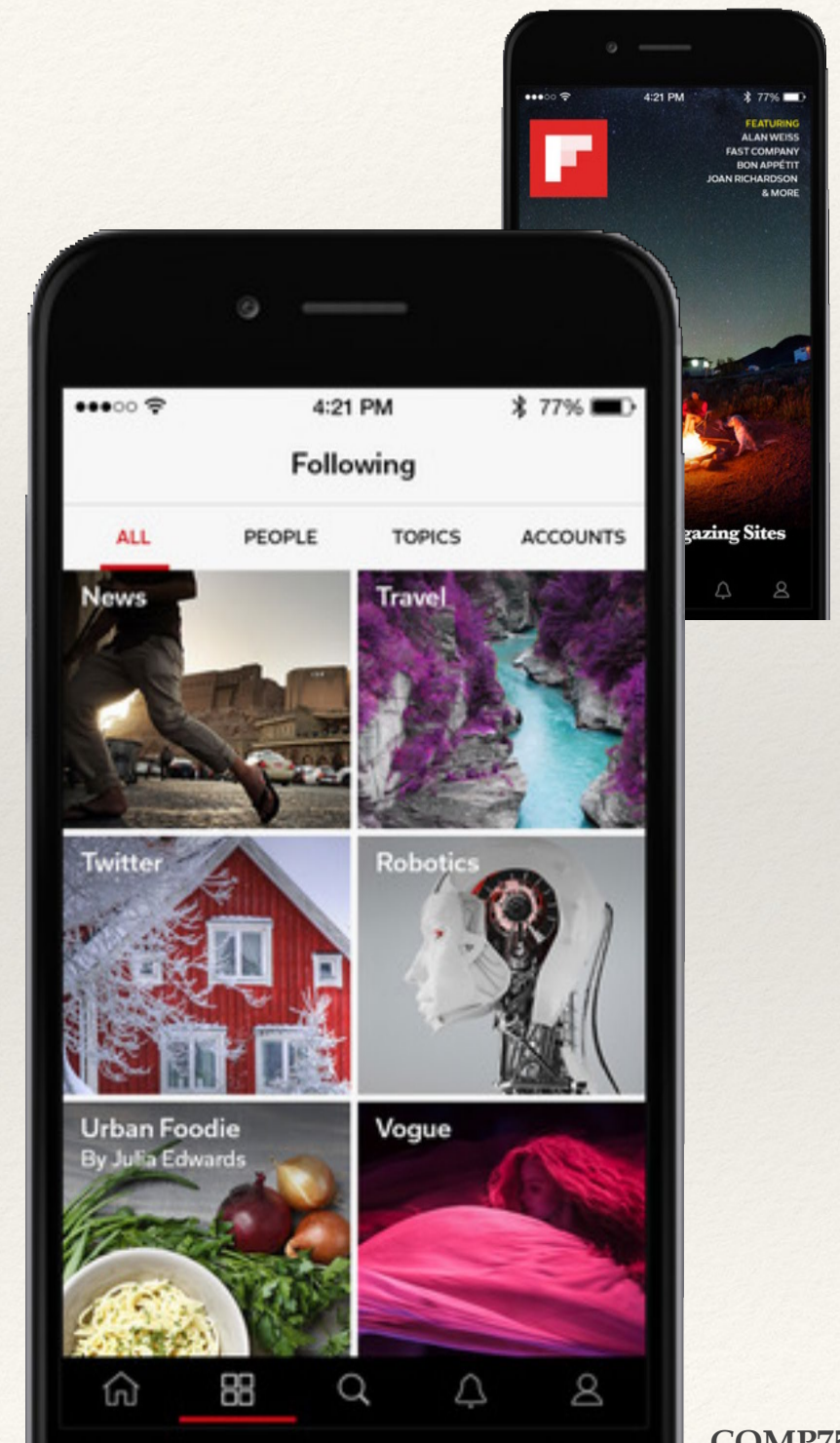


# Evolution of Documents (Cont'd)

- ❖ Evolve along three dimensions (Cont'd)

- ❖ *Dynamic Documents*

- ❖ During early days, the latency between an event and sharing information and experiences of the event is high
      - ❖ Newspapers were invented for reducing this latency between an event and its report
      - ❖ Television brought live events, but resource limitations kept this limited to only important events
    - ❖ Now, blogs, micro-blogs, and social media now bring real-time, dynamic updates and sharing live event information and experiences
    - ❖ Personalised dynamic documents can be designed to suit information relevance and personal needs



# Stages in Document Creation

## ❖ *Data acquisition and organisation*

- ❖ Identify the information, experiences, and message that need to be conveyed to users
- ❖ Need to factor in size limitations of documents
  - ❖ Due to attention span of user more than the physical requirements
- ❖ Metadata help organisation of data

| File Type     | Common metadata associated with the file                          |
|---------------|-------------------------------------------------------------------|
| Text Document | Name, Author, Length, Date Created, Date Last Modified, Type, ... |
| Photo         | Name, Capture Time, Compression Scheme, Exif, ...                 |
| Audio         | Sampling Rate, Number of Channels, Encoding, ...                  |
| Video         | Name, Creation Time, Compression, Length, Type, ...               |



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# Stages in Document Creation (Cont'd)

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## ❖ *Data Selection*

- ❖ The author selects the material from the content segments in the database considering the message he or she wants to convey
- ❖ Factors
  - ❖ Relevance of the segment to the message
  - ❖ Length of the segment
  - ❖ Media of the segment
  - ❖ How a segment could be combined with other segments
- ❖ Iterative process



# Stages in Document Creation (Cont'd)

## ❖ Editing

- ❖ It is the process of taking an existing document and modifying or refining it for its use in a given context

- ❖ Media dependent, require media-specific tools

- ❖ Text Editing

- ❖ Font, styles, colors, layout

- ❖ Photo Editing

- ❖ Highlight, addition, deletion of objects
  - ❖ Enhancement or restoration of visual characteristics

- ❖ Audio Editing

- ❖ Cut out, fade-in/out, copied, filtered

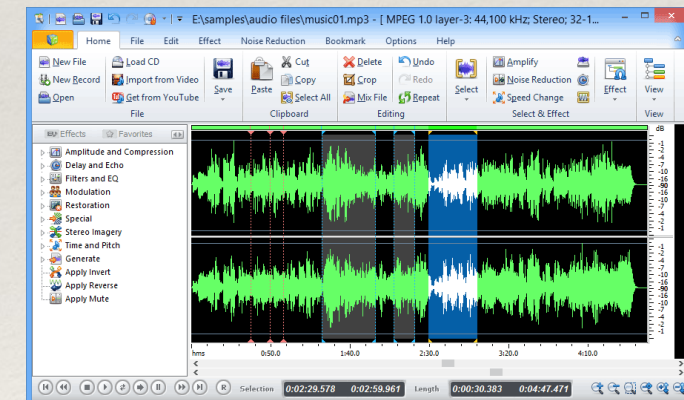
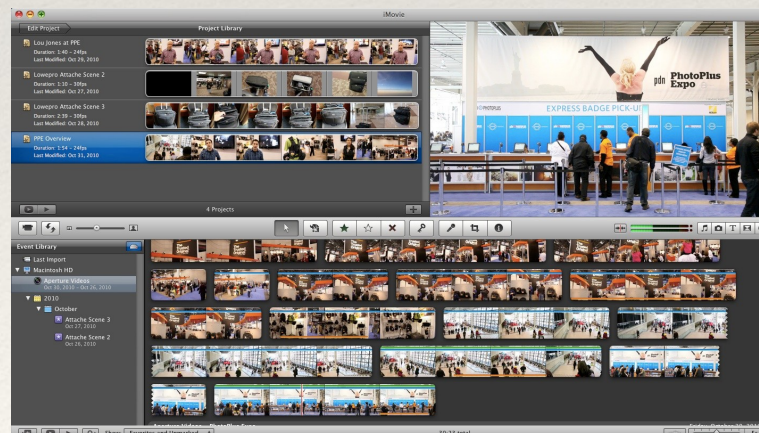
- ❖ Video Editing

- ❖ Modify objects like the operations in Photo Editing
  - ❖ Add, delete, video segments
  - ❖ Add text box (sub-title)
  - ❖ Add/Delete/Modify audio channels
  - ❖ Fade-in/out, filtering effect



Reference: [http://www.online-image-editor-clarkii.com/features/rich\\_text/](http://www.online-image-editor-clarkii.com/features/rich_text/)

Reference: <http://www.tadbirgaran.co/photoshoptuto/old%20to%20young.htm>



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# Multimedia Authoring Environment

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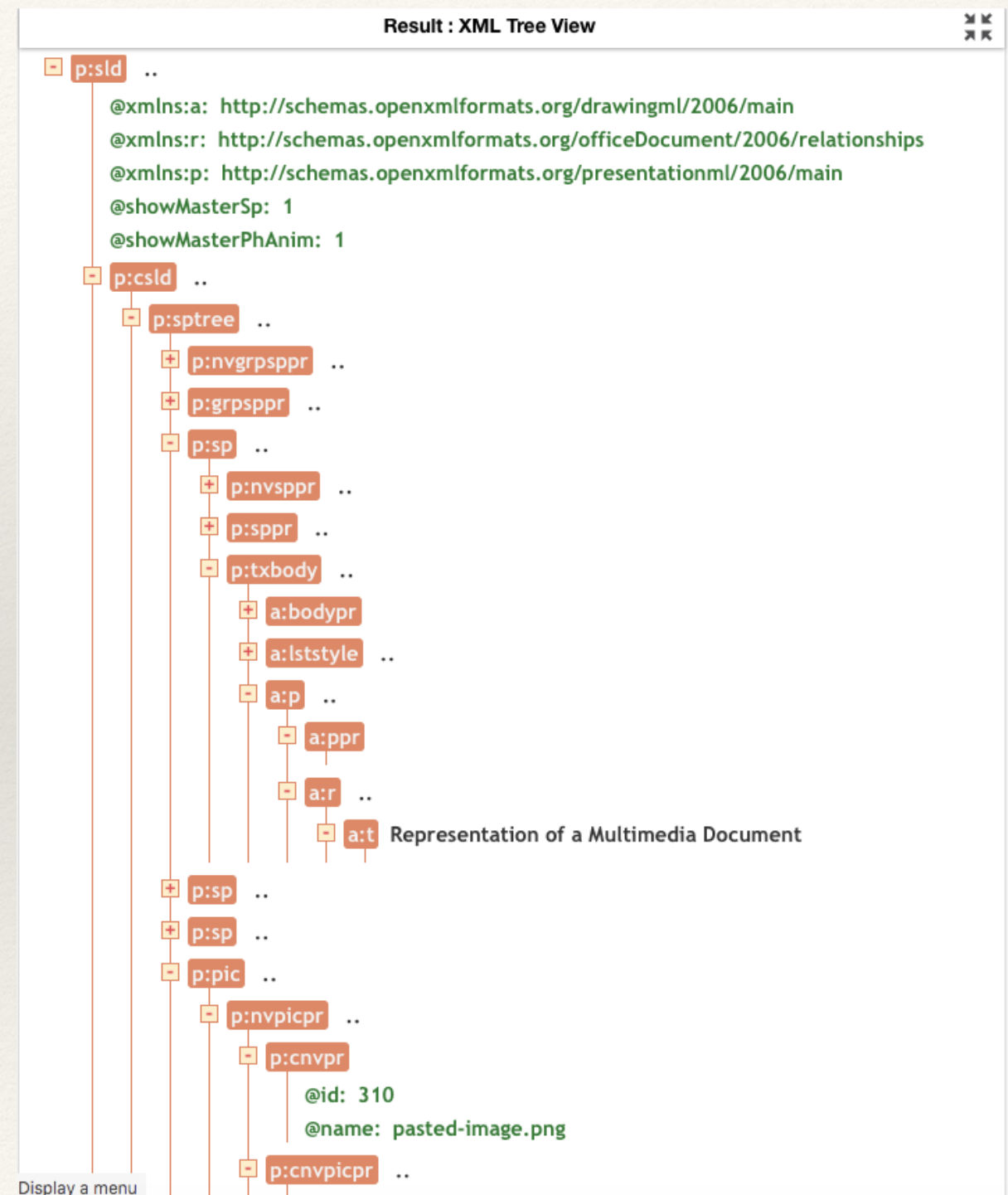
- ❖ Multimedia Authoring Environment should provide facilities for authoring each individual medium and to combine them effectively and efficiently
- ❖ Multimedia Authoring Environment should consider mechanisms for user interactivity as well
- ❖ Two fundamental aspects
  - ❖ Spatial composition of media assets
  - ❖ Temporal composition of media assets
- ❖ Different media assets have different spatial, temporal and other information attributes
  - ❖ These characteristics are usually stored as metadata alongside the data corresponding to the media
  - ❖ Authoring environment should take these metadata information into account



# Representation of a Multimedia Document

## ◆ Structure-based

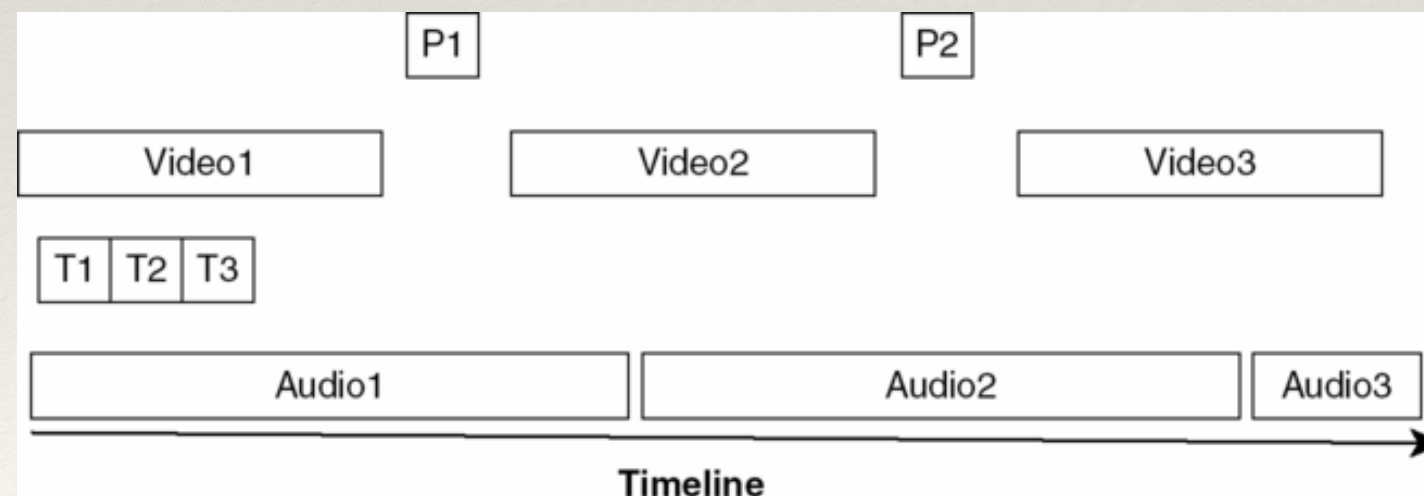
- ❖ Tree structure in which the root is a complete document and the leaf nodes are individual media elements or a pointer to those
- ❖ Intermediate nodes are “sub-documents” comprised of combinations of individual media elements
  - ❖ Composition rules and spatial and temporal layouts may be explicitly specified



# Representation of a Multimedia Document (Cont'd)

## ❖ Time-based Representation

- ❖ A multimedia document is organised around a timeline
- ❖ Different media elements are represented as different tracks synchronised with the master timeline
- ❖ Each track specifies which content element will appear during with the time interval
- ❖ One may also specify the relative spatial position of each media element



T: Text, P: Photo

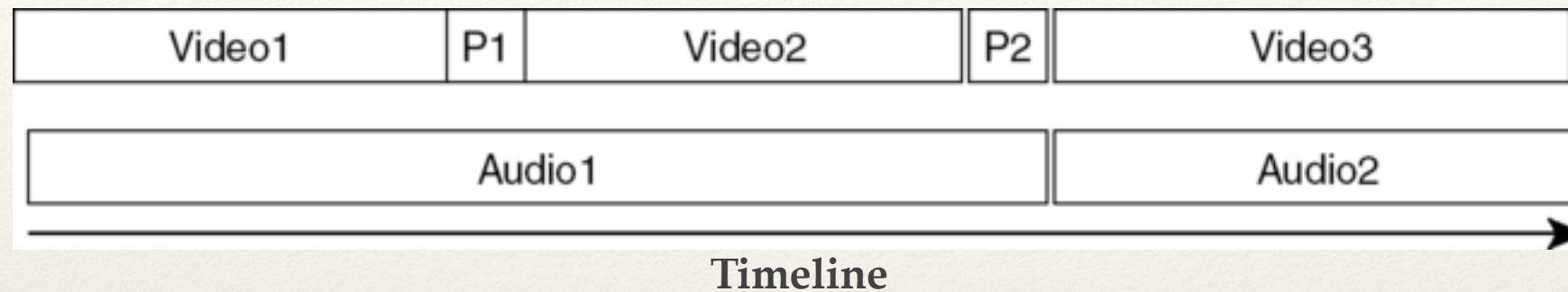


# Spatial Layout

- ❖ Multimedia documents can be rendered on screen, or in print
  - ❖ Screen has fixed spatial dimensions
    - ❖ 640x480, 1920x1080, 3840x2160, etc.
    - ❖ An author decides which media item should be displayed in which area of the screen
  - ❖ Printing require much higher dpi (dot per inch), and multimedia asset is preferred to be rendered at much higher resolution
    - ❖ Vector graphics is preferred



# Temporal Layout



- ❖ Multimedia content can be displayed as video/animation on screen
  - ❖ An authoring environment should specify different content content that will be used to constitute the video/animation
  - ❖ Tools should be provided to specify the time interval for the appearance, transitions, and disappearance of each content item on the screen
  - ❖ Location of the content on the screen could also be specified



# Synchronization

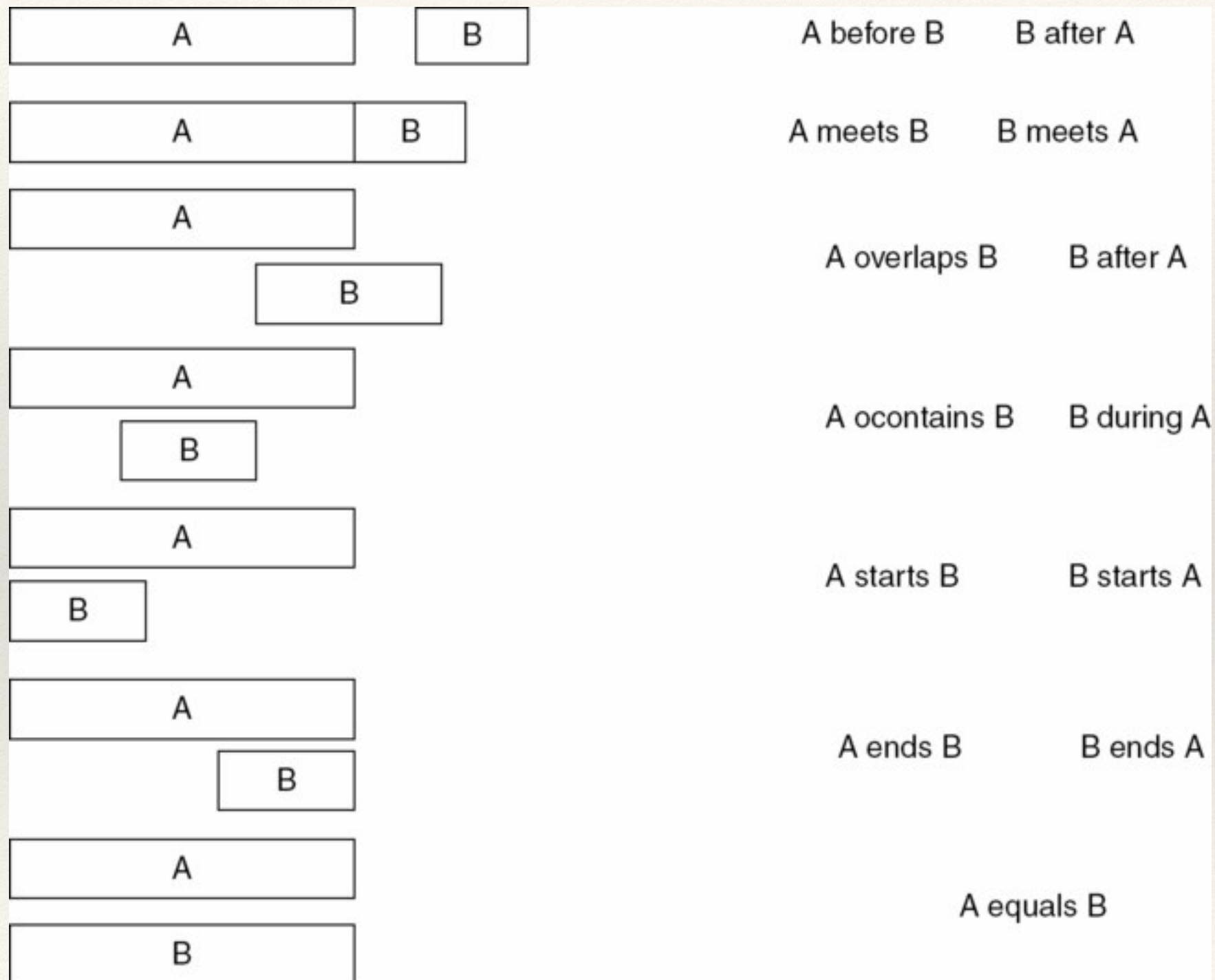
- ❖ For a multimedia document comprises different media contents of different types that are rendered in space and time
  - ❖ It is important to make sure that spatial and temporal relationships among different items are clearly specified by the author and are carried out by the system
- ❖ Synchronisation means establishing, specifying and then performing coordination of multiple media sources precisely in *space* and *time*



Source: [https://www.youtube.com/watch?v=tIW4\\_YibcVQ](https://www.youtube.com/watch?v=tIW4_YibcVQ)



# Temporal Relationships



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# Synchronization Levels in a System

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- ❖ Synchronisation has to be implemented at multiple levels in a system
  - ❖ Level 1
    - ❖ The lowest level, support must be provided to display each stream smoothly
    - ❖ Operating systems and lower communication layers are responsible for ascertaining a smooth single stream by bundling and making sure about the display requirements
    - ❖ They ensure minimal jitter at the presentation time of the stream
  - ❖ Level 2
    - ❖ Runtime support for synchronisation of multimedia streams and is implemented on top of level 1. This level makes sure that skews between various streams are bounded to acceptable limits dictated by applications
  - ❖ Level 3
    - ❖ This is the top level deals with the run-time support for synchronisation between time-dependent and time-independent media
    - ❖ This level is also responsible for handling of user interaction
    - ❖ The skews between time-dependent and time-independent media are within acceptable limits



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# Publishing Format

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- ❖ Documents are rendered on different sized screens
  - ❖ Computer Monitor
  - ❖ TV Screen
  - ❖ Mobile screens
- ❖ Bandwidth availability is also one of the concerns
  - ❖ Need adaption of the rendering process based on bandwidth
- ❖ Authoring systems should factor in the possibilities of presenting content under different rendering contexts

